

Future Energy Needs at a Glance

Team 201

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Introduction

As global energy demand rises, the need for sustainable, efficient, and innovative energy solutions has never been greater.

Why it Matters

- From powering economies to ensuring energy security, our future depends on how we address energy production, consumption, and technological advancements.

Our Focus

- This project provides a comprehensive look at future energy needs, exploring emerging trends, challenges, and solutions that will shape the global energy landscape.

Objective

- To analyze key factors influencing future energy demands and propose forward-thinking strategies that balance sustainability, affordability, and reliability.

Heilmeier Questions & Answers (1/2)

What are you trying to do? Articulate your objectives using absolutely no jargon.

- Develop a visual representation of projected energy demand by analyzing historical data from the past decade.

How is it done today; what are the limits of current practice?

- Existing forecasting models primarily rely on statistical techniques rather than machine learning, particularly for long-term predictions.

What's new in your approach? Why will it be successful?

- Instead of presenting static or generalized forecasts, our approach will incorporate geographic heat maps that dynamically illustrate predicted demand shifts over time, offering a more interactive and insightful visualization.

Who cares?

- Accurate regional forecasts are crucial for energy providers and policymakers, helping them prepare for future demands while minimizing blackouts, grid failures, and inefficiencies in energy distribution.

If you're successful, what difference and impact will it make, and how do you measure them (e.g., via user studies, experiments, ground truth data, etc.)?

- The project will enhance energy providers' ability to assess and meet future grid requirements, ultimately improving grid resilience. Success will be evaluated through biannual reviews of updated public energy data to validate the model's accuracy and effectiveness.

Heilmeier Questions & Answers (2/2)

What are the risks and payoffs?

- The model's effectiveness depends on the availability and consistency of high-quality data, which could vary as technology and data collection methods evolve. A widely adopted but later unreliable model could have negative consequences for energy providers and consumers. However, a robust and dependable forecasting model would significantly enhance energy efficiency and resource allocation.

How much will it cost?

- Since the model utilizes publicly available data, data access costs are negligible. However, making the model and visualizations accessible to the public may incur web hosting and server expenses, which will depend on the scale of deployment.

How long will it take?

- The development process is expected to take approximately one to two months, potentially less.

What are the midterm and final "exams" to check for success? How will progress be measured?

- By the midpoint, the model's structure and its integration with the visual and dynamic components of the user interface should be well-defined. By completion, a fully functional interface should be operational, displaying geographically mapped future energy demand. Progress will be monitored through team milestones outlined in the GANTT chart and by tracking interface development.

Summary of Literature Survey (1/2)

Electrical Energy Consumption in Hospitals (2016 – Long)

- **Main Idea:** Examines energy usage across hospital departments to anticipate future consumption.
- **Utility:** Helps predict local energy demand based on planned hospital construction and department configurations. Nationally, it reveals trends in hospital energy consumption.
- **Shortcomings:** Uses inconsistent data and is based on the German hospital system, limiting direct applicability to the U.S.

Electricity Consumption & Prediction Using Machine Learning Models

- **Main Idea:** Compares parametric and non-parametric ML models for energy forecasting and proposes a hybrid approach.
- **Utility:** Highlights machine learning's role in improving prediction accuracy.
- **Shortcomings:** Relies on discrete datasets without factoring in external influences like market conditions. Lacks geographic visualization, which is key to this project.

Energy Models for Demand Forecasting — A Review

- **Main Idea:** Surveys forecasting methods, from statistical techniques (e.g., ARIMA) to AI-driven models (e.g., neural networks, genetic algorithms).
- **Utility:** Provides a broad understanding of forecasting techniques and their evolution. Useful for selecting relevant models.
- **Shortcomings:** Outdated (published in 2012), missing recent advancements like deep learning. Lacks performance comparisons to determine the best approach.

Summary of Literature Survey (2/2)

Systematic Knowledge Trends for Building Energy Consumption (2021 – Long)

- **Main Idea:** A meta-review of energy consumption forecasting trends.
- **Utility:** Offers visualization and analysis techniques applicable to U.S. energy demand prediction.
- **Shortcomings:** Limited by field consensus and does not focus on industry- or region-specific predictors.

Methods of Forecasting Electric Energy Consumption: A Literature Review

- **Main Idea:** Reviews forecasting methods based on time horizon (short-term, medium-term, long-term) and compares statistical vs. machine learning approaches.
- **Utility:** Helps match forecasting techniques to specific use cases and lead times.
- **Shortcomings:** Qualitative analysis only—lacks empirical comparisons. Doesn't evaluate hybrid models or newer ensemble learning methods.

Energy Systems Modeling for 21st Century Challenges (2014 – Long)

- **Main Idea:** Discusses key challenges in energy modeling, including flexible demand, electrification, and renewable integration.
- **Utility:** Explains the importance of understanding future energy demand.
- **Shortcomings:** Lacks technical depth on models and datasets, making it a general overview rather than a standalone resource.

Expected Innovations

AI-Driven Forecasting Over Traditional Statistical Models

Current forecasting relies heavily on statistical techniques that lack adaptability to evolving energy trends. Our innovation integrates machine learning to enhance predictive accuracy, capturing long-term consumption patterns with improved flexibility.

Dynamic & Interactive Geographic Visualization

Unlike static models, our approach introduces geospatial heat maps to represent energy demand fluctuations over time. This allows stakeholders to visually assess demand shifts across regions, improving planning and decision-making.

Real-Time Adaptability to New Data

Existing models often fail to incorporate real-time or user-generated data. Our framework is designed to continuously refine predictions using the latest energy consumption metrics, policy changes, and infrastructure developments.

Enhanced Grid Resiliency & Planning Support

By providing accurate regional demand forecasting, our model helps utilities and policymakers anticipate energy needs, optimize grid operations, and reduce blackouts—key to improving overall grid stability and efficiency.

Cost-Efficient and Scalable Deployment

Leveraging publicly available data minimizes operational costs. The model's architecture is designed for scalability, allowing expansion from local to national levels with minimal infrastructure adjustments.

Measurable Success Through Iterative Evaluation

Progress will be assessed through biannual data reviews, tracking accuracy against real-world energy demand. Additionally, midterm checkpoints ensure continuous improvement of the model's performance and user interface integration.



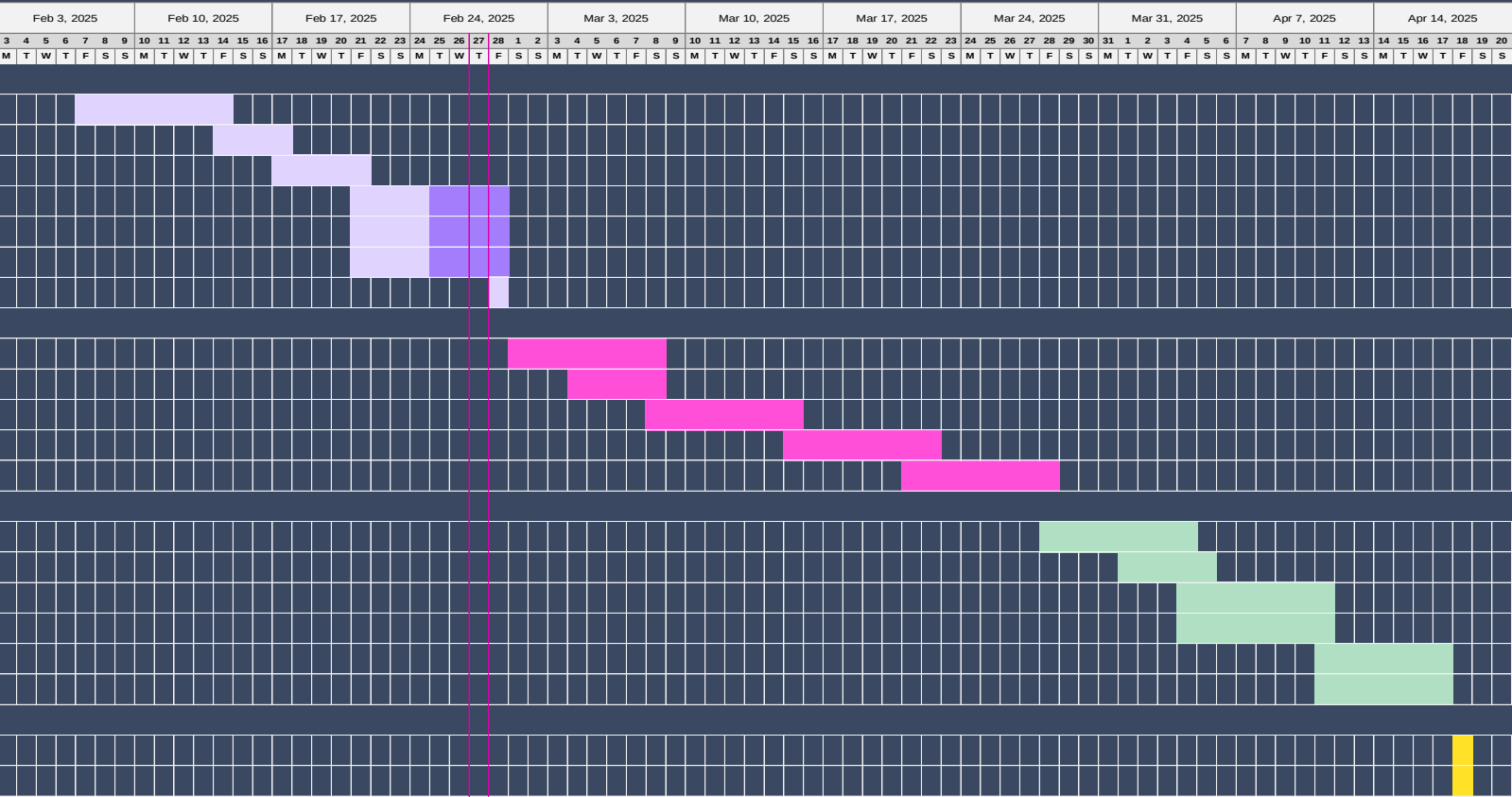
Plan for Activities

Forecasting Energy Demand

Team 202

Project start: **Fri, 2/7/2025**
Display week: **1**

TASK	ASSIGNED TO	PROGRESS	START	END
Project Proposal				
Generate Topic Ideas	Everyone	100%	2/7/25	2/14/25
Select Dataset	Everyone	100%	2/14/25	2/17/25
Gather Literature	Everyone	100%	2/17/25	2/21/25
Write Report	Sean/Trystan	50%	2/21/25	2/28/25
Draft Presentation	Ndubusi/Christopher	50%	2/21/25	2/28/25
Film Video	Hyung/Ravi	50%	2/21/25	2/28/25
Submit Proposal Materials	Everyone	100%	2/28/25	2/28/25
Approach and Minimum Viable Product				
Select Algorithm	Everyone		3/1/25	3/8/25
Clean Data	Everyone		3/4/25	3/8/25
Code MVP	Everyone		3/8/25	3/15/25
Evaluate MVP	Everyone		3/15/25	3/22/25
Progress Report	Everyone		3/21/25	3/28/25
Evaluation and Refinement				
Tune Model	Everyone		3/28/25	4/4/25
Finalize Results	Everyone		4/1/25	4/5/25
Build Visualization	Everyone		4/4/25	4/11/25
Design Poster	Everyone		4/4/25	4/11/25
Film Video	Everyone		4/11/25	4/17/25
Final Draft of Report	Everyone		4/11/25	4/17/25
Evaluation				
Submit Final Report	Gokce Aslan		4/18/25	4/18/25
Submit Video	Hayden Cook		4/18/25	4/18/25



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